

The data school_math.xlsx contains the following variables:

- **Classid**: class identifier
- **Childid**: student identifier
- **Schoolid**: school identifier
- **SES**: student's socioeconomic status
- **Housepov**: average household poverty by school
- **Mathprep**: class's amount of time spent on math preparation
- **sex**: male = 1
- **Mathknow**: class's mean math score
- **Yearstea**: years teacher has taught
- **minority**: minority = 1
- **mathkind**: math score in kindergarten
- **mathgain**: points gained in math score since kindergarten

These variables are measured at multiple levels. Use this data to answer the following questions.

Questions

- How many levels does the data have? Identify which variables are measured at each level.
- Using mathkind as the outcome, calculate the unadjusted intraclass correlation coefficients (ICCs) at all levels. Provide an interpretation of each ICC.
- Using the previous analysis, calculate the plausible values range of school-level mean kindergarten math scores.

For the remaining questions, use mathgain as the outcome.

- Calculate the unadjusted ICC at all levels. Provide an interpretation of each ICC.
- Which level-2 variable results in the largest decrease in the level-2 ICC when included in the model? Report the percent decrease.
- What is the change in the level-3 ICC when the level-3 variable is included? Report the percent decrease.
- Does the impact of student SES vary across schools? Fit a model that adjusts for student SES and test whether there is heterogeneity in the effect of SES at the school level.

$$\begin{aligned}
 \text{mathgain}_{ijk} &= \beta_{0jk} + \beta_{1j}\text{SES}_{ijk} + e_{ij} \\
 \beta_{0jk} &= \gamma_{00j} + b_{0jk} \\
 \gamma_{00j} &= \gamma_{000} + b_{0j} \\
 \beta_{1j} &= \gamma_{10} + b_{1j}
 \end{aligned}$$

where $b_{0jk} \sim N(0, \sigma_b^2)$ and $(b_{0j}, b_{1j}) \sim N(0, G)$.

(a) The data have three levels: student (level 1), class (level 2), and school (level 3). The variables on each level are:

- **Level 1:** SES, sex, minority, mathkind, mathgain.
- **Level 2:** Mathprep, Mathknow, Yearstea.
- **Level 3:** Housepov.

(b) The empty model can be run as follows:

```
proc mixed data = schooll covtest noclprint;
class schoolid classid;
model mathkind = / solution;
random intercept / subject = schoolid;
random intercept / subject = classid(schoolid);
run;
```

Here's the estimated variance of the random effects:

Covariance Parameter Estimates					
Cov Parm	Subject	Estimate	Standard Error	Z Value	Pr > Z
Intercept	schoolid	345.30	74.2841	4.65	<.0001
Intercept	classid(schoolid)	52.7594	49.7397	1.06	0.1444
Residual		1357.74	67.4918	20.12	<.0001

Class level ICC:

$$\frac{52.7594 + 345.30}{52.7594 + 345.30 + 1357.74} = 0.227$$

School level ICC:

$$\frac{345.30}{52.7594 + 345.30 + 1357.74} = 0.197$$

We can see that 22.7% of the variance is explained at the class level and 19.7% of the variance is explained by the school level. Interestingly adding the class level doesn't really add that much to the model.

(c) To get a plausible range of school level mean kindergarten math scores we will use

$$\beta_0 \pm 1.96 * \sigma_{school} = 465.35 \pm 1.96 * \sqrt{345.30} = (428.9, 501.7)$$

(d) Here's we'll switch to having mathgain as the outcome. Again, we'll run an unadjusted model.

```
proc mixed data = schooll covtest noclprint;
class schoolid classid;
model mathgain = / solution;
random intercept / subject = schoolid;
random intercept / subject = classid(schoolid);
```

`run;`

Here are the estimates of the random effect variances:

Covariance Parameter Estimates					
Cov Parm	Subject	Estimate	Standard Error	Z Value	Pr > Z
Intercept	schoolid	73.4249	34.7125	2.12	0.0172
Intercept	classid(schoolid)	116.77	46.8246	2.49	0.0063
Residual		1021.07	51.2525	19.92	<.0001

Class level ICC:

$$\frac{116.77 + 73.4249}{116.77 + 73.4249 + 1021.07} = 0.157$$

School level ICC:

$$\frac{73.4249}{116.77 + 73.4249 + 1021.07} = 0.061$$

We can see that 15.7% of the variance is explained at the class level and 6.1% of the variance is explained by the school level. For mathgain (as opposed to mathkind), there is a larger difference in the proportion of variability explained at the class versus school level. However, both ICC's are lower for this outcome.

- (e) One thing that we should do first, is to refit the unadjusted model on only those observations that have complete data on all level two variables (Mathprep, Mathknow, Yearstea). This way the sample size will stay consistent, which is important whenever comparing models. Here, these variables do not have any missing data, so we don't have to recalculate the ICC of the unadjusted model.

Here's the 3 models, the covariance parameter estimates, and their corresponding ICC's:

```
proc mixed data = school1_nomiss covtest noclprint;
class schoolid classid;
model mathgain = mathprep/ solution ddfm=betwithin;
random intercept / subject = schoolid;
random intercept / subject = classid(schoolid);
run;
```

Covariance Parameter Estimates					
Cov Parm	Subject	Estimate	Standard Error	Z Value	Pr > Z
Intercept	schoolid	68.9137	33.7862	2.04	0.0207
Intercept	classid(schoolid)	115.18	46.8466	2.46	0.0070

Covariance Parameter Estimates

Cov Parm	Subject	Estimate	Standard Error	Z Value	Pr > Z
Residual		1021.52	51.3515	19.89	<.0001

Class level ICC:

$$\frac{115.18 + 68.9137}{115.18 + 68.9137 + 1021.52} = 0.153$$

School level ICC:

$$\frac{68.9137}{115.18 + 68.9137 + 1021.52} = 0.057$$

```
proc mixed data = school1_nomiss covtest noclprint;
class schoolid classid;
model mathgain = mathknow/ solution ddfm=betwithin;
random intercept / subject = schoolid;
random intercept / subject = classid(schoolid);
run;
```

Covariance Parameter Estimates

Cov Parm	Subject	Estimate	Standard Error	Z Value	Pr > Z
Intercept	schoolid	90.0979	37.6961	2.39	0.0084
Intercept	classid(schoolid)	104.60	45.9646	2.28	0.0114
Residual		1018.90	51.0345	19.96	<.0001

Class level ICC:

$$\frac{104.60 + 90.0979}{104.60 + 90.0979 + 1018.90} = 0.160$$

School level ICC:

$$\frac{90.0979}{104.60 + 90.0979 + 1018.90} = 0.074$$

```
proc mixed data = school1_nomiss covtest noclprint;
class schoolid classid;
model mathgain = yearstea/ solution ddfm=betwithin;
random intercept / subject = schoolid;
random intercept / subject = classid(schoolid);
run;
```

Covariance Parameter Estimates

Cov Parm	Subject	Estimate	Standard Error	Z Value	Pr > Z
Intercept	schoolid	71.2066	34.6341	2.06	0.0199
Intercept	classid(schoolid)	119.11	47.2576	2.52	0.0059

Covariance Parameter Estimates

Cov Parm	Subject	Estimate	Standard Error	Z Value	Pr > Z
Residual		1021.03	51.2405	19.93	<.0001

Class level ICC:

$$\frac{119.11 + 71.2066}{119.11 + 71.2066 + 1021.04} = 0.157$$

School level ICC:

$$\frac{71.2066}{119.11 + 71.2066 + 1021.04} = 0.059$$

None of the variables result in a notable decrease in the ICC. In fact, two variables actually raise the ICC! Overall, the mathprep variable results in the biggest decrease, but it's hardly any.

The percent decrease is: $(0.157 - 0.153)/0.157 * 100 = 2.5\%$.

- (f) Now we'll fit a model with just Housepov, the only level 3 variable. Here are the random effect variances.

Covariance Parameter Estimates

Cov Parm	Subject	Estimate	Standard Error	Z Value	Pr > Z
Intercept	schoolid	76.3989	35.2267	2.17	0.0150
Intercept	classid(schoolid)	116.43	46.7943	2.49	0.0064
Residual		1020.85	51.2335	19.93	<.0001

Class level ICC:

$$\frac{116.43 + 76.3989}{116.43 + 76.3989 + 1020.85} = 0.159$$

School level ICC:

$$\frac{76.3989}{116.43 + 76.3989 + 1020.85} = 0.063$$

Again, there is very little difference in the level 3 ICC. It is actually higher when adjusting for the level 3 variable.

The percent decrease in the school level ICC is: $(0.061 - 0.063)/0.061 * 100 = -3.3\%$. Here the percent decrease is negative, because it's actually an increase.

- (g) We can answer these questions with a two or three level model. The difference is if we are going to have a random effect at the class level or not.

If we look at it as a two-level model, we'll fit the following (here j is the school):

$$\begin{aligned} \text{mathgain}_{ij} &= \beta_{0j} + \beta_{1j}SES_{ij} + e_{ij} \\ \beta_{0j} &= \gamma_{00} + \gamma_{01}\text{housepov}_j + b_{0j} \\ \beta_{1j} &= \gamma_{10} + \gamma_{11}\text{housepov}_j + b_{1j} \end{aligned}$$

If we look at it as a three-level model, we'll fit the following (here j is the school and k is the class):

$$\begin{aligned} \text{mathgain}_{ijk} &= \beta_{0jk} + \beta_{1j}SES_{ijk} + e_{ij} \\ \beta_{0jk} &= \gamma_{00j} + b_{0jk} \\ \gamma_{00j} &= \gamma_{000} + \gamma_{001}\text{housepov}_j + b_{0j} \\ \beta_{1j} &= \gamma_{10} + \gamma_{11}\text{housepov}_j + b_{1j} \end{aligned}$$

We'll fit both models and see which has better fit statistics.

Here's the code for the two-level model (no class level random effect):

```
proc mixed data = school1_nomiss covtest noclprint;
class schoolid classid;
model mathgain = housepov ses housepov*ses / solution ddfm=betwithin;
random intercept ses / subject = schoolid;
run;
```

Fit Statistics

-2 Res Log Likelihood	10685.8
AIC (Smaller is Better)	10689.8
AICC (Smaller is Better)	10689.8
BIC (Smaller is Better)	10695.1

Here's the code for the three-level model:

```
proc mixed data = school1_nomiss covtest noclprint;
class schoolid classid;
model mathgain = housepov ses housepov*ses / solution ddfm=betwithin;
random intercept ses / subject = schoolid;
random intercept / subject = classid(schoolid);
run;
```

Fit Statistics

-2 Res Log Likelihood	10676.7
AIC (Smaller is Better)	10682.7
AICC (Smaller is Better)	10682.8

Fit Statistics

BIC (Smaller is Better) 10690.7

We can see that the three-level model appears to fit better (likelihood ratio test confirms this). Let's look at the estimates of the random effect variances from this model:

Estimated G matrix is not positive definite.

Covariance Parameter Estimates

Cov Parm	Subject	Estimate	Standard Error	Z Value	Pr > Z
Intercept	schoolid	77.2259	35.5478	2.17	0.0149
ses	schoolid	0	.	.	.
Intercept	classid(schoolid)	116.75	46.8415	2.49	0.0063
Residual		1021.83	51.3245	19.91	<.0001

Notice the "Estimated G matrix is not positive definite" warning. Clearly the ses random effect is not needed in the model. This indicates that the effect of ses doesn't vary across schools.